TÉCNICO
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Performance Evaluation and Dimensioning of Networks and Systems

METI

Lab 2 Report [EN]

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1 Introduction

This report presents the solutions to the three exercises assigned to class 5 in Laboratory 2 of the Performance Evaluation and Dimensioning of Networks and Systems (DDRS) course. It follows the objective–approach–results–conclusion structure used in the Laboratory 1 report.

2 A. Discrete-time Markov chains

2.1 Bernoulli and 2-DTMC sample paths

Objective: Compare Bernoulli processes with two-state discrete-time Markov chains (2-DTMCs) using two parameter pairs that have the same stationary mean but markedly different temporal behavior.

Approach: The supplied helper `dtmc_bernoulli_plot.R` uses the transition matrix

$$P = \begin{bmatrix} \alpha & 1-\alpha \\ \beta & 1-\beta \end{bmatrix},$$

The stationary probability of state 1 is

$$\pi_1 = \frac{1-\alpha}{1-\alpha+\beta}.$$

For $(\alpha, \beta) = (0.9, 0.1)$, the helper generates 200 DTMC samples and an independent Bernoulli process with success probability $\pi_1 = 0.5$, plots both paths, and saves them together as `p3.jpeg`.

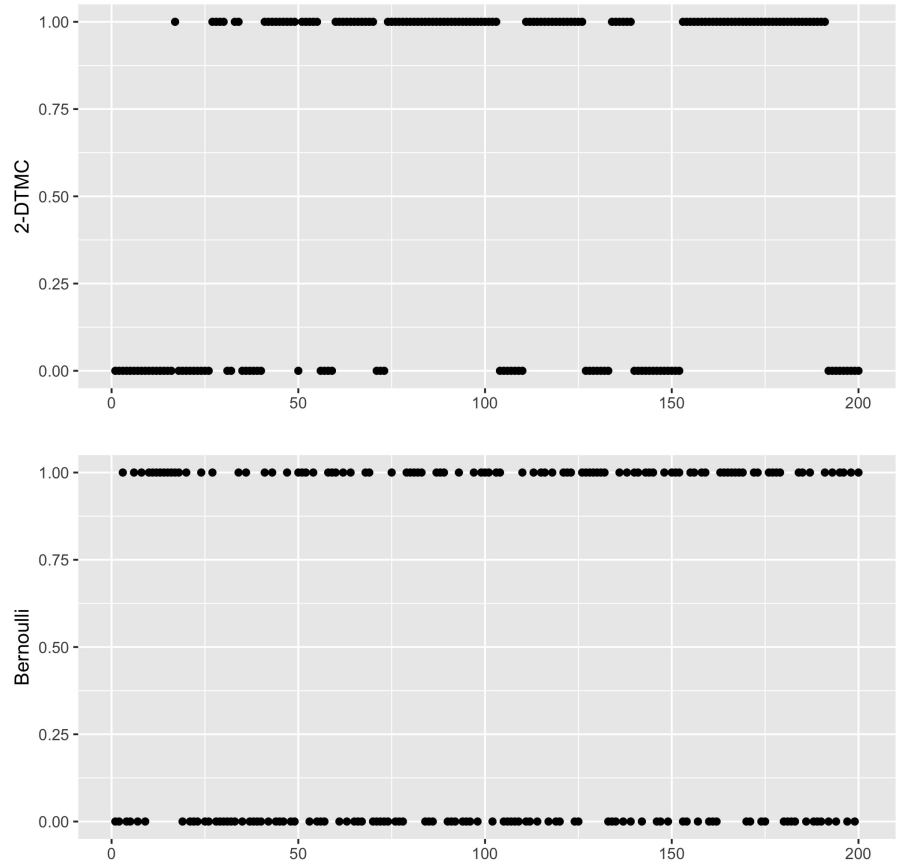


Figure 1: Sample paths from the 2-DTMC and Bernoulli process generated by the supplied helper.

Results and conclusion: Since $\alpha = 0.9$, the DTMC tends to remain in its current state, producing longer runs than the independent Bernoulli process. Both processes have stationary mean 0.5, but only the DTMC has temporal dependence. This illustrates why a Markov model can represent clustering in a sequence that an independent Bernoulli model cannot.

2.2 Estimating transitions from the attack trace

Objective: Estimate the transition probabilities of a 2-DTMC from the sequence of daily server attack indicators in `2dtmcdata.txt`.

Approach: The script `E2_AttackDTMC.R` imports the supplied file with the tab separator specified in the exercise. It counts each consecutive transition $i \to j$ and divides by the number of transitions originating in state i. The trace contains 2,000 observations and thus 1,999 transitions.

Table 1: Estimated transition counts and probabilities for the attack trace.

	Transition counts		Estimated probabilities	
Current state	To 0	To 1	To 0	To 1
0 (not attacked)	623	577	0.5192	0.4808
1 (attacked)	577	222	0.7222	0.2778

Results and conclusion: The estimated transition matrix, with rows denoting the current state, is

$$\widehat{P} = \begin{bmatrix} 0.5192 & 0.4808 \\ 0.7222 & 0.2778 \end{bmatrix}.$$

The fitted chain estimates an attack probability of approximately 0.3997 in steady state. An attack is followed by another attack with estimated probability 0.2778; otherwise, the process returns to the non-attacked state with probability 0.7222. These are empirical estimates from this finite trace, rather than known properties of all server attacks.

2.3 Slotted ALOHA

Objective: Study the theoretical and simulated throughput of slotted ALOHA with a finite number of users, explain the effect of the parameters N, σ, and p, and compare ALOHA with carrier-sensing protocols.

2.3.1 Theoretical throughput

Approach: The supplied `aloha_plot.R` script uses `aloha_theo()` to plot theoretical throughput versus σ. With $N = 10$, it plots four curves for p: 0.3, 0.4, 0.5, and 0.6. With $N = 25$, it plots $p = 0.2$ and $p = 0.3$. In both plots, σ is logarithmically spaced as specified in the script.

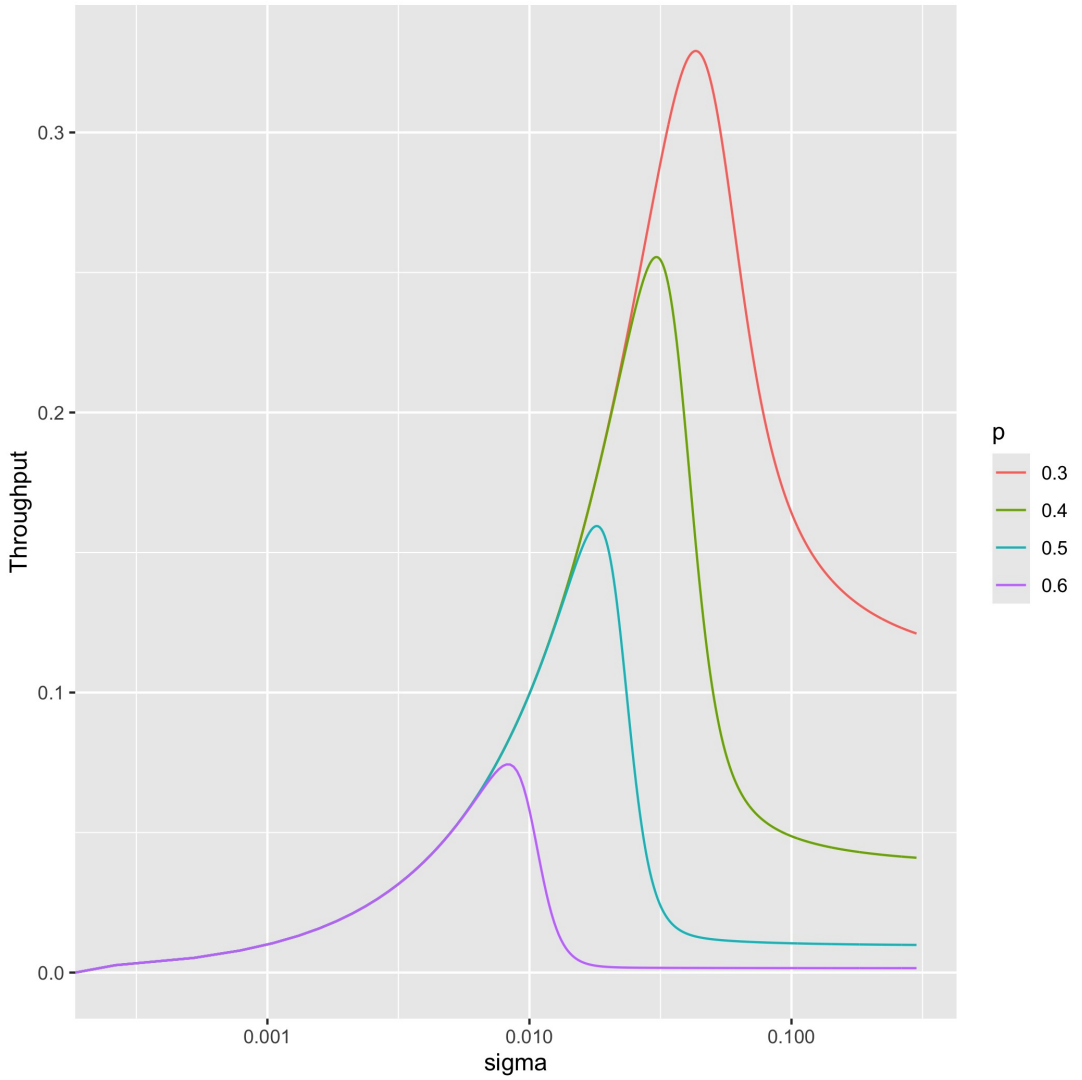


Figure 2: Theoretical slotted-ALOHA throughput for $N = 10$ and the four requested values of p.

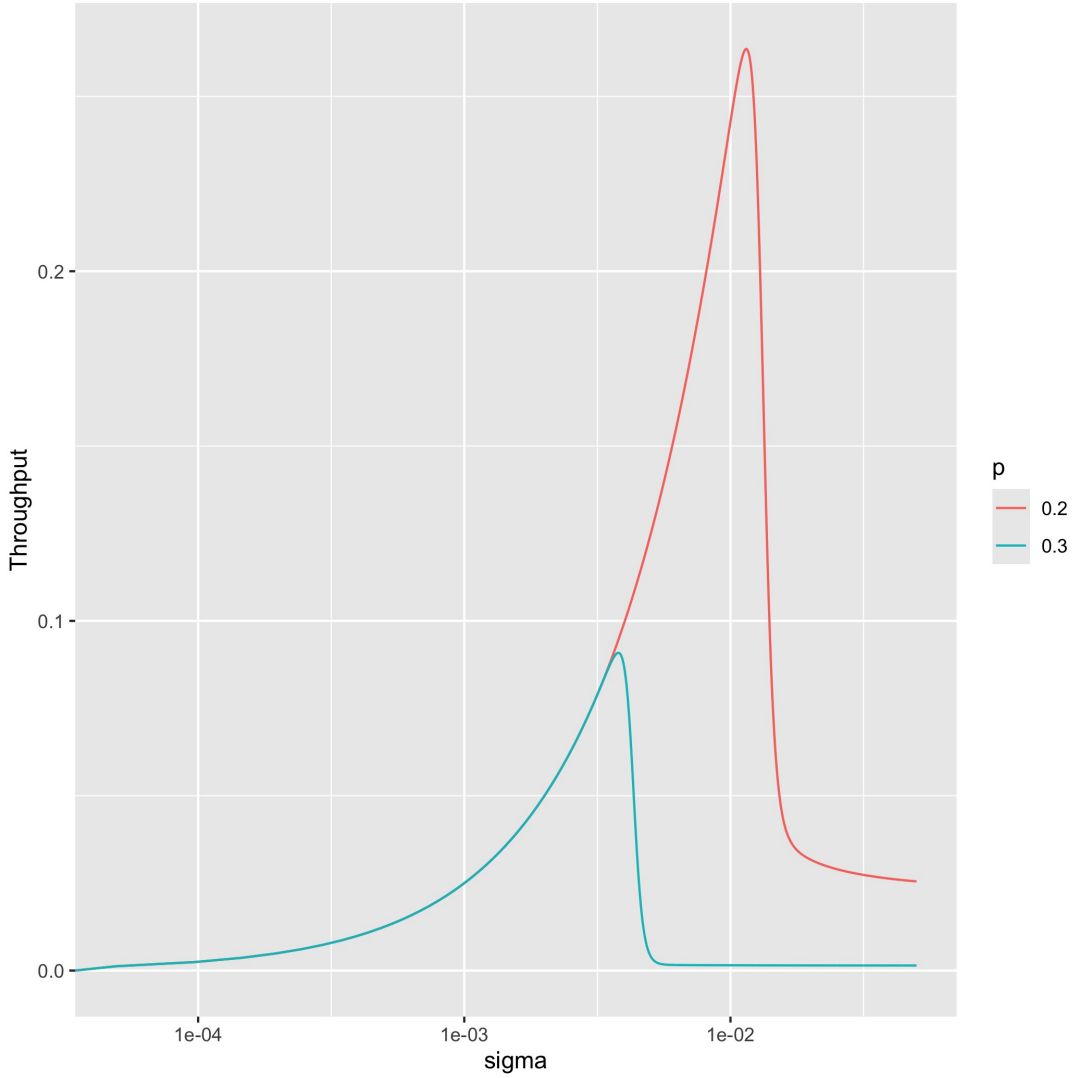


Figure 3: Theoretical slotted-ALOHA throughput for $N = 25$ and the two requested values of p.

Results and conclusion: Increasing σ makes more THINKING users transmit. Throughput can first increase as successful attempts become more frequent, then decrease when the added attempts create more collisions and BACKLOGGED users. Increasing p makes BACKLOGGED users retry more often; this can clear a small backlog, but can also increase collisions when many users are waiting. Increasing N adds potential transmissions and contention, so throughput depends on the interaction between N, p, and σ, rather than increasing monotonically with any one parameter.

2.3.2 Discrete-event simulation

Approach: The same script runs one slot at a time. Each user attempts with the probability associated with its state. If exactly one user transmits, the slot is successful and a successful backlogged user returns to THINKING. If two or more transmit, there is a collision and each transmitting THINKING user becomes BACKLOGGED; existing backlogged users remain so.

After a warm-up of 100,000 slots, throughput is estimated over 1,000,000 slots. The theoretical reference is computed by `aloha_theo()`.

Table 2: Theoretical and simulated throughput for three operating points.

N	σ	p	Theoretical	Simulation
10	0.001	0.3	0.01000	0.01004
10	0.100	0.3	0.16415	0.16309
10	0.100	0.6	0.00160	0.00157

Results and conclusion: The simulated estimates are close to their theoretical values for all three configurations. The results illustrate the sensitivity to the offered load and retransmission probability: the middle setting has the highest throughput, while raising p to 0.6 at the same σ leaves more users contending in the backlog and substantially reduces successful transmissions.

2.3.3 ALOHA vs. CSMA, CSMA/CD and CSMA/CA

Approach: Random-access throughput is limited by collisions. The protocols differ in (i) how long a frame is exposed to collisions (the *vulnerable period*) and (ii) how much channel time a collision wastes. Let T be the frame transmission time, τ the propagation delay and $a = \tau/T$.

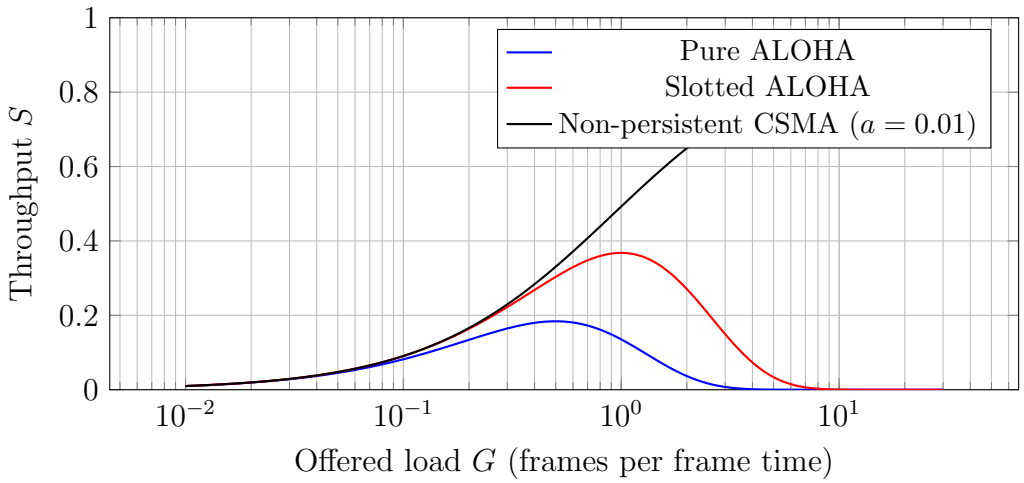


Figure 4: Classical throughput curves (infinite population).

Table 3: Random-access protocols compared.

Protocol	How it works	Throughput	Used in
Pure ALOHA	Transmit immediately; no ACK ⇒ retry after a random delay.	Vulnerable period $2T$. $S_{\max} = 1/(2e) \approx 0.18$.	ALOHAnet, LoRaWAN uplink
Slotted ALOHA	Same, but only at slot boundaries.	Vulnerable period T. $S_{\max} = 1/e \approx 0.37$.	RFID (EPC Gen2), cellular/satellite random-access channels
CSMA	"Listen before talk": if the channel is busy, defer (1-persistent: send when idle; non-persistent: wait a random time and re-sense; p-persistent: send with probability p per slot).	Vulnerable period shrinks to τ. For $a = 0.01$, non-persistent reaches ≈ 0.8; falls as a grows.	Early packet radio; basis of the two variants below
CSMA/CD	CSMA plus listening *while* sending. On a collision: abort, send a jam signal, wait a random time (binary exponential backoff). Needs frame time $\geq 2\tau$.	A collision wastes $\approx 2\tau$ instead of T. Efficiency $\approx 1/(1+5a)$ (approx.).	Shared Ethernet (IEEE 802.3, hubs/coax); not in switched full-duplex
CSMA/CA	Wait for the channel to be idle (DIFS), then count down a random backoff, frozen while busy; receiver ACKs; optional RTS/CTS. Avoids collisions since a radio cannot detect them while sending.	Fewer collisions than CSMA, but backoff and ACK overhead lower the gain.	IEEE 802.11 Wi-Fi, IEEE 802.15.4 (Zigbee)

Conclusion: Each step wastes less channel time. Slotting halves the vulnerable period ($2T \to T$) and doubles the peak throughput. Carrier sensing cuts it to about τ, so for $a \ll 1$ throughput exceeds 0.8. Collision detection reduces the cost of a collision from a full frame to about 2τ. CSMA/CA gives up part of this gain to work where collisions cannot be detected (wireless). When a is large (e.g. satellite links), sensing is of little help and ALOHA-type access remains in use. The simulated peaks in Section X agree with the slotted-ALOHA limit $1/e$.

References: Bertsekas & Gallager, *Data Networks*, Ch. 4; Rom & Sidi, *Multiple Access Protocols*; Kleinrock & Tobagi (1975); Kurose & Ross, *Computer Networking*.

3 B. Discrete-time Markov chains and Continuous-time Markov chains

3.1 4

3.2 5

4 C. Queuing systems

4.1 6

4.2 7

4.2.1 .a

4.2.2 .b

4.2.3 .c

5 D. Workloads

6 Conclusion

The three class 5 exercises demonstrate how DTMCs extend independent Bernoulli models by representing temporal dependence, how transition probabilities can be estimated directly from observed state changes, and how the stationary behavior of a contention protocol can be studied analytically and by simulation. The estimated and simulated results are consistent with the corresponding models, while also showing that finite data and operating parameters matter when interpreting performance.